

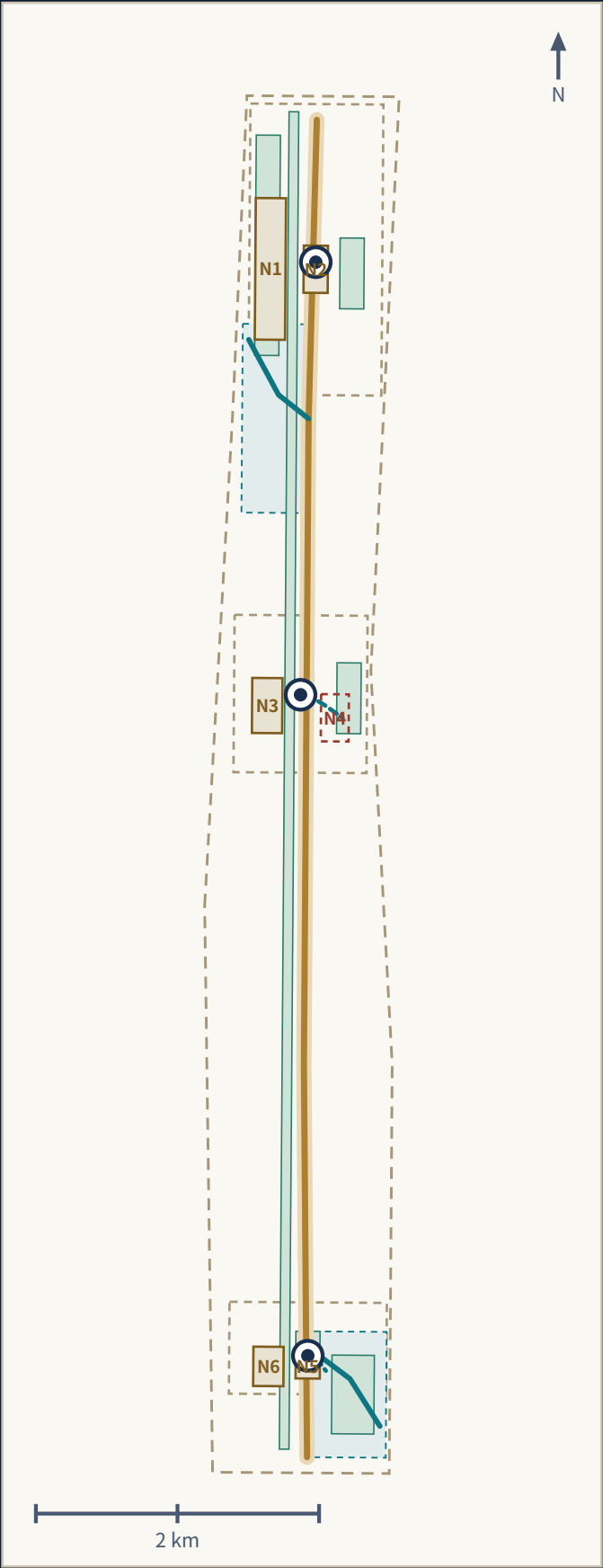
HAIDIAN AI AXIS

FROM COAL TO COMPUTE

Centennial Jing-Zhang AI Innovation Belt international urban design open call • open-source concept proposal by an agent team

Heritage spine → three-station roles → two-wing supply/feedback → six spatial prototypes → ten operating scenarios → one annual ledger

Document	Page • 14 pages
Submission	submissions/harry-water/haidian-ai-axis
Package state	See current manifest.json
Metrics	14 current recalculations • 23 establishment paths
Output chain	proposal • figures • HTML • A3/A0 • machine checks



A single reading spine: every chapter is one link on it

- 1

Heritage spine
First a continuous, free, exitable public space
- 2

Three-station roles
Exchange / verify / launch as a differentiated node sequence
- 3

Two-wing supply and feedback
Factor supply and real-demand stress testing
- 4

Six spatial prototypes
A combinable grammar of gateway, courtyard, station, market, garden, stitch
- 5

Ten operating scenarios
Every card has human review, an offline alternative and exit conditions
- 6

One annual ledger
Checks back whether the first five links still hold

Four core propositions of Haidian AI Axis

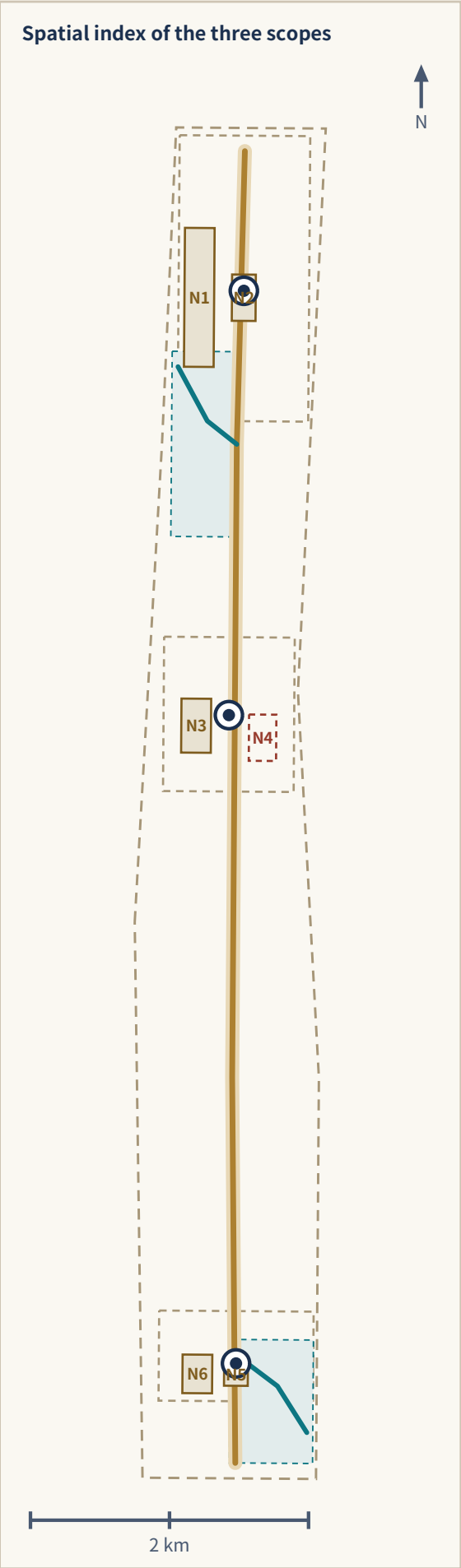
- 1
- Build a distinctly Haidian innovation-space sequence: continuous spine, ordered nodes, layered time and public sharing.
- 2
- Tell a century of regional exchange: Jing-Zhang–Ping-Sui–Jing-Bao evolution links energy, knowledge and compute collaboration.
- 3
- Establish a Haidian–Ulanqab twin-city compute model that assigns edge, local and nearby capacity by task type.
- 4
- Create humane human–AI service through digital, physical and staffed access calibrated by operating tests.

Four evidence tiers

Tier	Content	Design role	Deepening route
E1 Official task basis	Open-call announcement, agent taskbook	Three scopes, three key areas, design tasks and output depth	Connect to official boundaries and technical appendices
E2 Public standards	Urban design measures, regulatory plan measures, land-use classification guide	Urban-design depth, public-space requirements, land-use semantics	Translate into project controls during professional design
E3 Public primary data	Government statistics, bulletins, policy documents, official reporting	Regional climate and compute-policy background, historical facts	Calibrate with site surveys, agreements and operating data
E4 Project-side material	Middle-East case project sites and host-institution releases	Mechanism reference, operating organization	Evaluate applicability through Haidian pilot data

Submitted evidence-layer index

File	Features / content
geometry/site_boundary.geojson	SITE-001
geometry/key_areas.geojson	PROV-KEY-001 • PROV-KEY-002 • PROV-KEY-003
geometry/land_use.geojson	LU-001 • LU-002 • LU-003 • LU-004 • LU-005 • LU-006
geometry/buildings.geojson	BLDG-ST01-001 • BLDG-ST02-001 • BLDG-ST03-001
geometry/roads.geojson	ROAD-AXIS-001 • ROAD-WING-TECH-001 • ROAD-WING-XIAOYUE-001 • ROAD-STITCH-N4 • ROAD-STITCH-N5
geometry/green_space.geojson	GREEN-AXIS-001 • GREEN-QINGHE-001 • GREEN-XIAOYUE-001 • GREEN-ST01-001 • GREEN-ST02-001 • GREEN-ST03-001
geometry/public_space.geojson	N1 • N2 • N3 • N4 • N5 • N6
geometry/constraints.geojson	CONSTRAINTS • WING-TECH-001 • WING-XIAOYUE-001
geometry/phasing.geojson	PHASE-001 • PHASE-002 • PHASE-003
metrics.json	14 current recalculations • 23 establishment paths
sources.json / assumptions.json	42 sources • 18 assumptions



Three scopes: announced extent, work objective and output

Scope / size	Announced extent	Work objective	Position on the spine	Output
Coordinated research area approx. 43.6 km²	North: North 5th Ring Rd; East: Jingzang Expwy; South: Xizhimenwai St; West: Wanquanhe Rd	Energy–compute–knowledge network north of Beijing, industry ecosystem and mechanism set	Regional background of the spine; origin of two-wing supply and feedback	Strategy diagram, mechanism chain, scenario and operating framework
Overall design area approx. 11.4 km²	Urban and industrial areas 1–2 km around the Jing-Zhang Heritage Park	Heritage spine, six spatial prototypes, renewal framework and character control	The heritage spine and the six spatial prototypes	Land use, roads, green, public space, buildings and phasing layers
Key detailed-design areas approx. 368.4 ha	North to south: Zhongzhiyuan 192.1 ha, AI Origin Community 104.3 ha, Dazhongsi 72.0 ha	Detailed design of three differentiated urban stations and their concept nodes	Where the three-station roles and ten scenarios land	Positioning, structure, renewal, active travel, public space, AI scenarios and deepening focus

The three levels implement one spine at increasing depth

The coordinated research area asks what the corridor exchanges in the AI era; the overall design area turns exchange into walkable and lingerable space; the key areas let three stations carry exchange, validation and launch.

The current working extent supports concept positioning, structural comparison and synchronized recalculation. Official polygons trigger one coordinated update of areas, ratios, building action extents and planning controls.

Working base: SITE-001 and PROV-KEY-001/002/003 are maintainer-registered provisional rough surfaces (official_boundary=false, boundary_precision=provisional_rough). Thin dashed lines show the current concept extent; all nine layers and related metrics update together when official polygons are published.

From coal to compute: one northbound corridor, three eras of functional transformation

Era	What the northbound corridor carries	What it gives Beijing / Haidian	What Haidian exports
Railway era	Coal, goods, people	Industrial fuel, goods exchange, regional linkage	Manufactured goods, markets and urban services
Zhongguancun era	Talent, knowledge, equipment, capital	Research and industrial innovation capacity	Technology outputs, firms and services
AI era	Compute tasks, models, data products, services	Suitable nearby green compute and elastic capacity	Algorithms, models, standards, talent, scenarios and demand

This is a continuing two-way exchange: northern resources move south while Haidian's algorithms, models, standards, talent and scenario demand move north. The 1909 Jing-Zhang line linked Beijing and Zhangjiakou; the later Ping-Sui–Jing-Bao evolution extended this exchange toward Inner Mongolia.

Beijing Central Axis: three distinctive correspondences

What is distinctive	Correspondence	Contemporary Haidian expression
A composite of urban buildings and heritage sites	The spine is a family of spaces	Modern public space and linear wayfinding
Nodes differ in function and order, balanced in pairs	A differentiated node sequence with paired balance	Functional complementarity and equitable access
Construction and everyday life of different eras keep overlaying	Continuous overlay of time and culture	Heritage, innovation and community life evolving together

Three local planning actions

Action	Spatial and operating action	Delivery focus
1. Cross-axis stitching	Stitching across the railway, Qinghe and Xiaoyue blue-green cross-links, community transfer and east-west public space; spine and cross-links matter equally	Select the form through transport, accessibility and engineering studies
2. Public sharing	Keep the spine and landmarks open through digital, physical and staffed service	Create a stable opening mechanism through ownership, maintenance and service-window agreements
3. Long-term governance	From one-off construction to annual operation and review: the Axis Timetable, an open test season, the compute ledger, human review, public feedback and annual updates	Use the pilot charter to establish organization, budget, audit and public participation

Delivery pathway for Haidian–Ulanqab twin-city compute

Public network reporting provides Beijing-direction fibre and an approximately 4.2 ms link figure; the project agreement confirms Haidian links, application latency and service level.

Facility-specific PUE and natural-cooling data start the green-compute study; the project ledger uses partner-confirmed facility, period and metering definitions.

The Haidian–Ulanqab memorandum provides a cooperation basis; the next stage creates a project agreement for tasks, capacity, price, SLA, budget and incident responsibility.

Task tiers assign computation: field control and sensitive tasks use edge/local systems, low-latency service uses local/near-edge, ordinary inference and peaks follow the agreement, and training/batch work uses suitable nearby resources.

Privacy boundary

Outbound tasks use authorized, classified and minimized data; identity, biometrics, minors', health and movement information stays protected locally.

Energy, carbon and water ledger

Per task, record latency, queueing, failure rate, energy, the applicable PUE definition, green-power share, carbon, water and switchover time, each with its definition and source. Start with test specifications and table structures, then integrate real operating data.

Failover and annual review

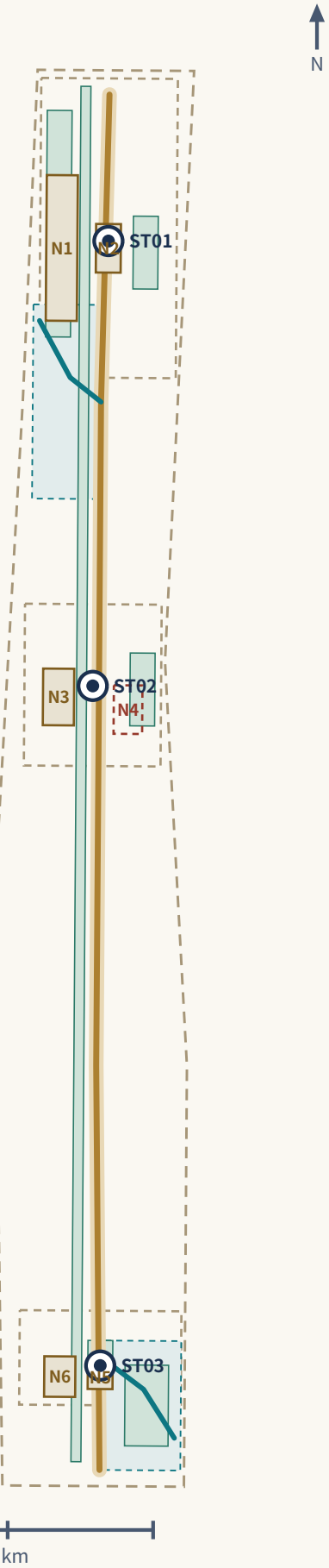
Every switchover retains a clear human decision point; human takeover is a standard resilience process; every switchover and takeover enters the public ledger. One open test season and ledger review per year is published and can revise the task-tiering rules in return.

One axis, three stations, two wings

The spine is first a free, continuous, accessible, climate-adapted public space and only then a display interface

HAIDIAN AI AXIS: FROM COAL TO COMPUTE
Railway Heritage. AI Innovation. Everyday Life.

One spine, three stations, two wings



ST01 • Exchange and dispatch Zhongzhiyuan AI Independent Innovation Acceleration Area

Compute Exchange

Positioning
Full-stack independent innovation and AI-governance capability; the new axis’ s place for deciding and explaining where tasks are computed.

Spatial structure
A Qinghe-front public living room organized as river → garden → courtyard → R&D cluster; dispatch and testing gather in public-facing central courtyards.

AI scenarios
02 Safety Governance Sandbox, 06 Jing-Zhang Heritage and AI Public Education, 08 Haidian–Ulanqab Green Compute Dispatch Test, 10 Climate Digital-Twin Operations Test.

Concept nodes
N1 Qinghe Compute Commons — Waterfront linear park + a group of open courtyards
N2 Fail-safe Test Yard — Hard-surfaced yard that can be closed or opened + outer gallery

ST02 • Verification and incubation Beijing AI Origin Community

Origin Commons

Positioning
Validation and incubation for a world-class AI ecosystem; public services receive community and university review before citywide use.

Spatial structure
Courtyard-centered small-scale network linking campus, park, and neighborhood; open-source honors line the slow-mobility spine.

AI scenarios
01 Open-Source Launch Hall, 04 Newcomer 72-Hour Settling-In, 06 Jing-Zhang Heritage and AI Public Education, 09 Research-to-SME Conversion Test.

Concept nodes
N3 Origin Courtyard — Semi-enclosed community courtyard + open street-level ground floors
N4 Campus–Station Stitch — A continuous walking-and-cycling stitch with option comparison

ST03 • Launch and translation Dazhongsi AI Industry Cluster

Axis Launch Station

Positioning
Launch and conversion area for intelligence-native business; the axis’ s public, market, and international interface.

Spatial structure
Market-led, with four-quadrant pedestrian continuity and ground-floor services around rail stations; launch and cultural content follow the walking network.

AI scenarios
05 Barrier-Free Continuous Passage, 07 Safe Route Home for Night Workers, 03 Network/Power Fail-safe, 10 Climate Digital-Twin Operations Test.

Concept nodes
N5 Four-Quadrant Crossing — Continuous walking surfaces and waiting spaces at four corners
N6 New Axis Launch Market — Adaptable open street frontage + a permanent canopy

Six urban spaces: a repeatable, combinable block grammar

Space type	Urban role	Typical design action	Layer
Gateway	Information, accessibility and public-service interface	Station entrances, wayfinding, accessible ramps and a staffed desk in authorized windows	N5
Courtyard	Exchange, community courses, rest and small trials	Semi-open courtyard, bookable test bays, shade and seating	LU-004
Station	Exchange node for compute, mobility, service or events	Edge compute, transfer and event gathering with parallel digital and non-digital entrances	ROAD-AXIS-001
Market	Showcase, procurement, culture and everyday commerce	Open street frontage, activated ground floor, temporary exhibits	LU-003
Garden	Shade, winter sun, stormwater detention and quiet rest	Climate public space, floodable green, rain gardens	GREEN-AXIS-001
Stitch	East-west links across rail, road and water barriers	Active-travel stitching and ecological links; feasibility studied separately	ROAD-STITCH-N4

The six spaces are combinable prototypes for the professional teams that follow. The spine begins as free, continuous, accessible and climate-adapted public space, then layers wayfinding, service and smart facilities to strengthen movement, shade, rest and accessibility.

Two wings and the coordination loop

Zhongguancun technology-service wing
Factor supply: capital, legal, IP, talent and international networks

Xiaoyue River scenario-enablement wing
Real-demand feedback: stress-testing community, ecology, mobility and everyday needs

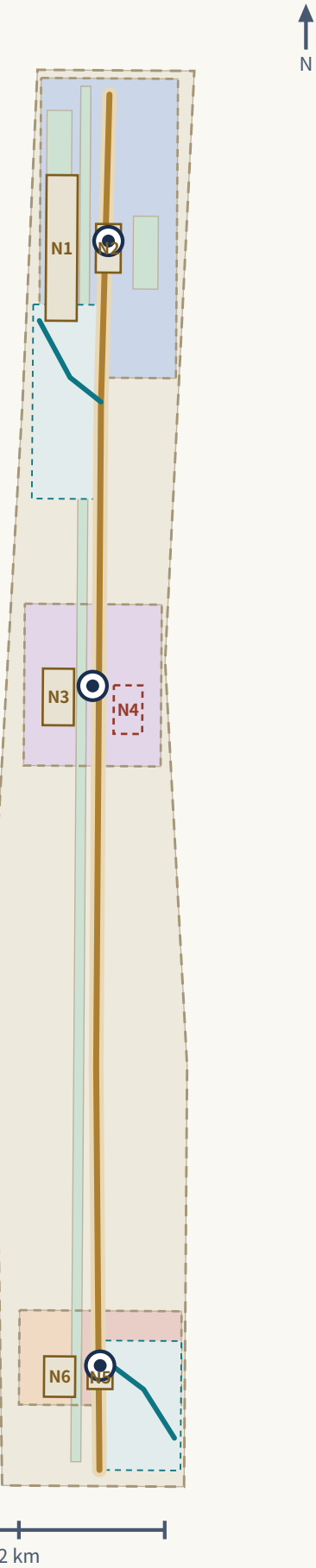
Loop direction: Origin Community verifies → Zhongzhiyuan exchanges and dispatches → Dazhongsi launches and translates → the two wings supply factors and real demand → back to Origin Community for the next round.

Three positionings
Centennial Jing-Zhang cultural belt, urban AI everyday-life belt, AI convergence innovation belt.

Five functions
A full-stack independent AI innovation system, a world-class AI innovation ecosystem, a new AI+ scenario-enablement paradigm, an intelligent and lively AI city, and a global voice in AI governance.

Localized mechanism chain
Climate public space → research and talent anchor → a standing operating body → twin-city green compute cooperation → a digital twin with a rights boundary. The order is itself the design judgement: make the public space good without any device first, then layer the smart systems on.

Conceptual land-use partition



Conceptual land-use partition (closed, gapless, non-overlapping; areas recomputed in EPSG:4548)

Feature	Name	Code	Area	Derivation
LU-001	AI R&D innovation land (diagrammatic concept)	0802	144.2 ha	PROV-KEY-001 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with
LU-002	Community service facility land (diagrammatic concept)	0702	88.2 ha	PROV-KEY-002 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with
LU-003	Commercial and enterprise service land (diagrammatic concept)	05	23.3 ha	PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the market-node (N6) side is retained; clipped to
LU-004	Cultural / railway heritage interpretation land (diagrammatic concept)	0803	31.2 ha	PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the heritage/crossing-node (N5) side is retained;
LU-005	Park and open space land (diagrammatic concept)	1401	129.5 ha	Union of all six submitted green-space concept polygons (GREEN-AXIS-001, GREEN-QINGHE-001, GREEN-XIAOYUE-001, GREEN-ST01-001, GREEN-ST02-001, GREEN-ST
LU-006	Reversible reserve / flexible land (diagrammatic concept)	16	724.9 ha	SITE-001 minus the union of LU-001..LU-005; computed and recomputed in EPSG:4548 with a shared 1mm snapping grid so shared boundaries with adjacent la

Regulatory depth in three tiers

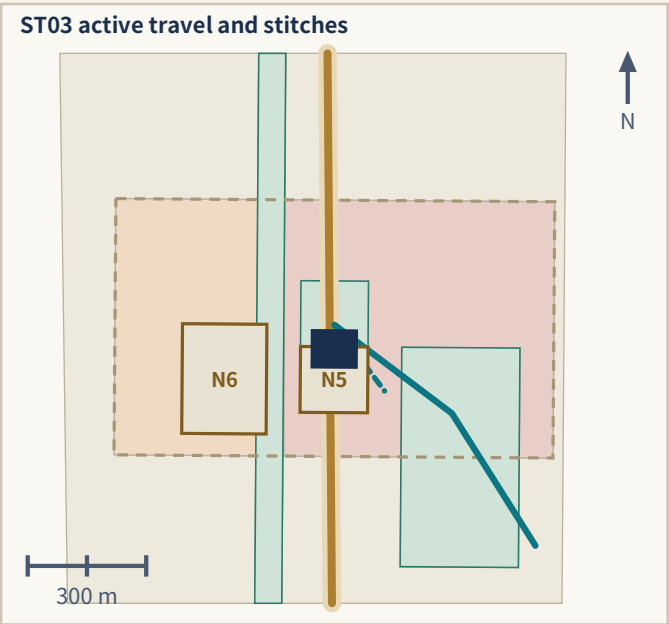
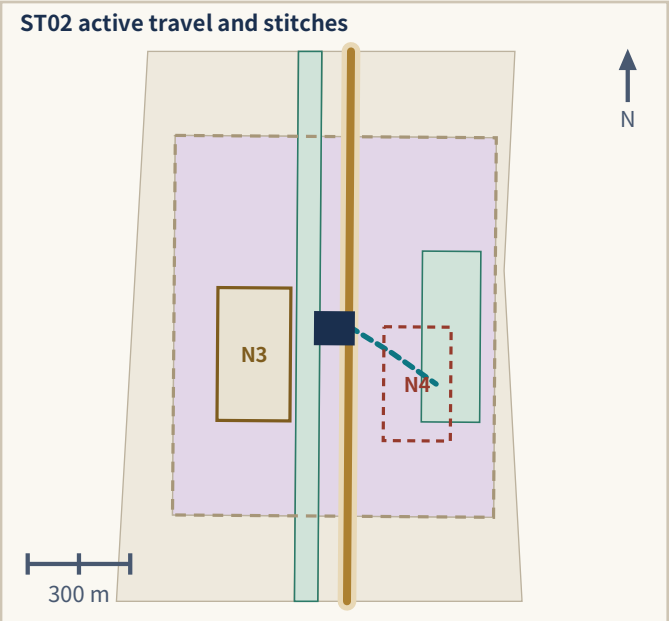
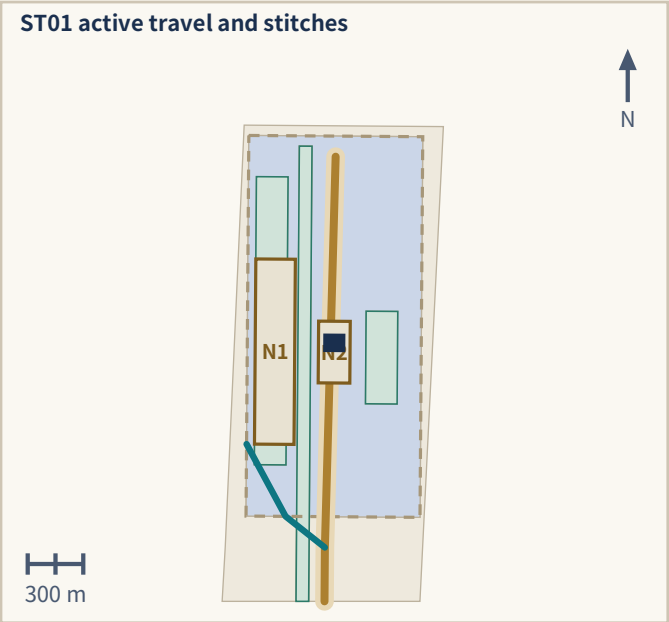
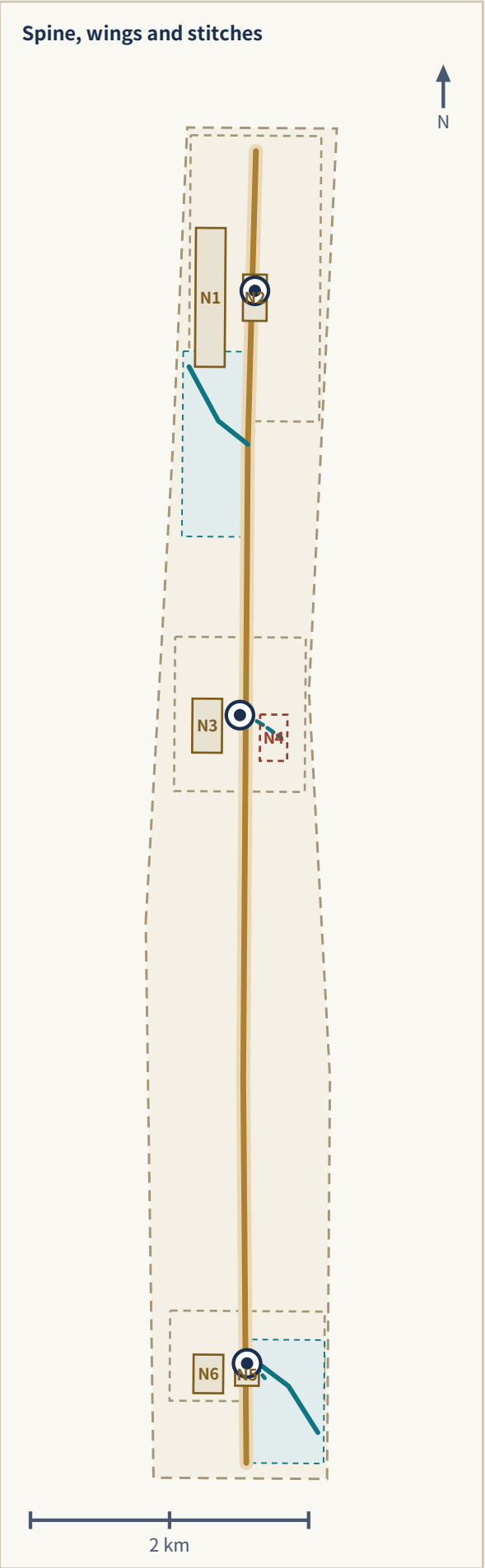
- (1) Directly recomputable from the submitted geometry**
Land-use structure, green and public-space ratios, building footprint area. All are based on the provisional boundary and are for discussion only.
- (2) Controls requiring an official regulatory plan**
FAR, building height, density, setbacks, rights-of-way and facility standards. All are marked unknown or pending official conditions this round; no speculative value is passed off as an approved control.
- (3) Performance indicators needing operating data**
AI innovation index, talent density, active-travel accessibility, scenario usage and event participation enter annual review rather than becoming planning conclusions.

Four-tier building strategy and concept intervention envelopes

Tier	Design action	Precondition
Retain	Keep massing and frontage; maintenance only	Condition survey
Light renovation	Activate ground floor, open frontage, tidy roof, add accessibility	Ownership and use consent
Reversible addition	Removable, relocatable lightweight structures	Fire, structural and planning permits
New build pending verification	Position intent only; no scale or height	All regulatory, ownership, utility and traffic conditions confirmed

Feature	Name	Footprint
BLDG-ST01-001	Zhongzhiyuan Compute Exchange (conceptual envelope)	11,945.1 m²
BLDG-ST02-001	AI Origin Community Origin Commons (conceptual envelope)	11,949.8 m²
BLDG-ST03-001	Dazhongsi Axis Launch Station (conceptual envelope)	11,957.0 m²
Total	building_footprint_area_sqm	35,851.926 m²

The three envelopes carry survey_status=pending_survey and are new concept intervention volumes, not existing footprints, GFA or FAR; floor_area_ratio remains unknown. No parcel-level retain/renovate/demolish conclusion is given.



Five conceptual alignments (lengths recomputed in EPSG:4548; not engineering alignments)

Feature	Name	Class	Nodes	Length
ROAD-AXIS-001	North-south public greenway spine (concept)	greenway	N1 N2 N3 N4 N5 N6	9,439 m
ROAD-WING-TECH-001	Zhongguancun technology-service wing link (concept)	pedestrian	N1 N2	714 m
ROAD-WING-XIAOYUE-001	Xiaoyue River scenario-enablement wing link (concept)	greenway	N5 N6	768 m
ROAD-STITCH-N4	Campus-Station Stitch connector (concept)	transit_connection	N4	306 m
ROAD-STITCH-N5	Four-Quadrant Crossing stitch (concept)	pedestrian	N5	210 m
Total	conceptual_mobility_length_m			11,437.434 m

N4 Campus–Station Stitch research task

N4 Campus–Station Stitch is a dedicated research task. Transport, accessibility and engineering studies will compare crossing, end-space and operating options to form a safe, continuous and legible public link between campus and station.

Technical record: design_only_closed_pending_study; the opening scheme follows the completed study.

Utilities and new infrastructure: turning invisible compute into sited, accountable urban facilities

Edge compute points

Placed in station-type spaces for low-latency public services; always co-located with rest, enquiry and charging, never as standalone machine rooms.

Climate nodes

Combined with gardens and courtyards for shade, warmth, drinking water and temporary storm refuge; the physical carrier of the extreme-weather modes.

City-side expression of failover

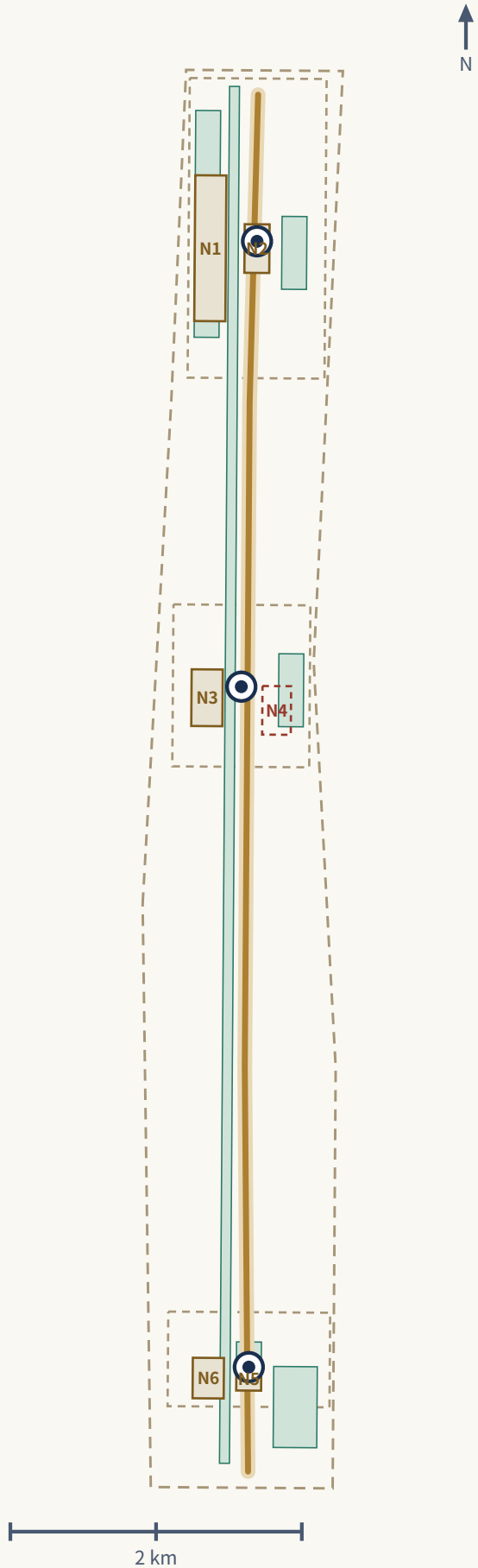
When an external link fails, stations display a degraded state and switch to staffed service; a black screen with nobody to ask is not acceptable.

Conventional utilities

Existing water, drainage, power, telecom and sanitation capacity is unknown and is listed as a precondition for formal development; no load calculation is offered.

The CONSTRAINTS surface is the professional coordination interface for the next planning, mobility, utility, ecology and heritage inputs.

Blue-green network and heritage spine



Six blue-green concept surfaces (recomputed areas)

Feature	Name	Area
GREEN-AXIS-001	Heritage spine continuous shade corridor (concept)	64.5 ha
GREEN-QINGHE-001	Qinghe riverside detention green belt (concept)	26.5 ha
GREEN-XIAOYUE-001	Xiaoyue River seasonal refuge green space (concept)	16.6 ha
GREEN-ST01-001	Zhongzhiyuan station commons green (concept)	8.5 ha
GREEN-ST02-001	AI Origin Community station commons green (concept)	8.5 ha
GREEN-ST03-001	Dazhongsi station commons green (concept)	4.7 ha
Union	green_ratio = 0.113434	129.5 ha

Six public-space prototype nodes (recomputed areas)

Node	Name	Prototype	Area
N1	Qinghe Compute Commons	Garden + Courtyard	21.3 ha
N2	Fail-safe Test Yard	Courtyard + Station	5.7 ha
N3	Origin Courtyard	Courtyard + Gateway	8.3 ha
N4	Campus–Station Stitch	Stitch + Station (research task)	6.5 ha
N5	Four-Quadrant Crossing	Gateway + Station	2.8 ha
N6	New Axis Launch Market	Market + Courtyard	5.9 ha
Union	public_space_ratio = 0.044371		50.6 ha

A climate-first public spine: everyday activity, storm detention, safe recovery

Everyday mode

Continuous shade, ventilation, seating and rest/cooling points; establish coverage through walking-distance control.

Storm-detention mode

Activate tiered storage, floodable green and rain gardens; set capacity through hydrology and engineering data.

Safe-recovery mode

Run desilting, inspection and reopening; human collaboration is a standard resilience process.

Three AI pilgrimage landmarks and cultural narrative (concept)

Compute Exchange (Zhongzhiyuan)

Centred on a publicly readable dispatch and ledger interface that turns “where a task is computed” into visitable, questionable public content; the honour display shows public test records and evaluation reports, not company rankings.

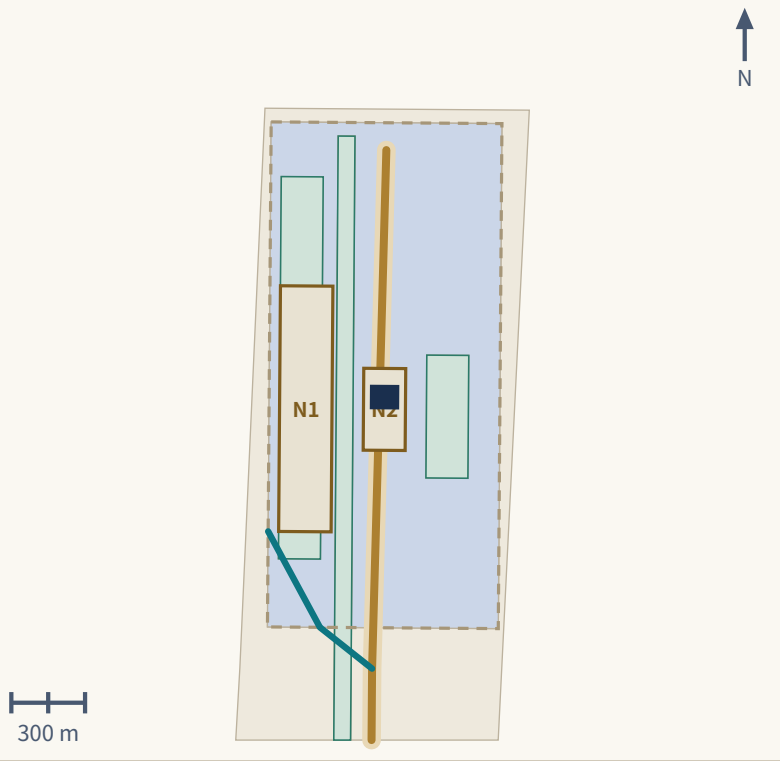
Origin Commons (AI Origin Community)

Centred on an open-source honour wall, community oral history and live model verification, recording contributors and contribution events; any display can be removed at a contributor's request.

Axis Launch Station (Dazhongsi)

Centred on the annual Axis Timetable and result launches while doubling as an everyday urban living room; international exchange and local everyday use share the same space.

ST01 concept spatial index



ST01 • Exchange and dispatch
Zhongzhiyuan AI Independent Innovation Acceleration Area

Compute Exchange
PROV-KEY-001 • announced approx. 192.1 ha • recomputed 192.9 ha

Positioning
Full-stack independent innovation and AI-governance capability; the new axis’ s place for deciding and explaining where tasks are computed.

Spatial structure
A Qinghe-front public living room organized as river → garden → courtyard → R&D cluster; dispatch and testing gather in public-facing central courtyards.

Building renewal
Retention and light renovation first; activate ground floors and roof equipment levels; use reversible additions for testing and display and calibrate new space with planning and ownership conditions.

Mobility and public space
Separate external traffic from internal walking/cycling, build a continuous Qinghe path, and keep staffed and physical service entrances at interchanges. Gardens support shade, stormwater, and quiet rest.

AI scenarios
02 Safety Governance Sandbox, 06 Jing-Zhang Heritage and AI Public Education, 08 Haidian-Ulanqab Green Compute Dispatch Test, 10 Climate Digital-Twin Operations Test.

Deepening focus
Coordinate river, flood, energy, safety, and ownership conditions and establish time- and zone-based public access for testing.

Project hooks
HX-02 Qinghe-Xiaoyue blue-green cross-link; HX-04 Compute Exchange and public ledger interface

N1 Qinghe Compute Commons

清河算力公园

Garden + Courtyard S06 S08 S10

Opportunity
Turn the Qinghe edge into a public living room where compute becomes visible, understandable, and discussable.

Spatial action
A continuous linear park with open courtyards, public-readable dispatch and ledger interfaces, and shaded walks; a Garden + Courtyard prototype.

Operating modes
Daytime waterfront rest and viewing; lower night lighting with continuous wayfinding and callable human help; courtyards as shade/rain shelter and low sections as floodable green during storms.

Users
R&D and evaluation staff, residents, walkers, cyclists, and visitors.

Delivery inputs
River and flood controls, ownership and maintenance, frontage-renovation consent, energy and safety requirements.

Deepening route
Fix location, scale, flood design, ownership, and operation from the agreed prototype.

Manual / offline collaboration (declared geometry field)
Printed/static signage is on-site; a staffed help point is a design target during authorized service periods only, subject to confirmed operator, and is not a permanent-staffing claim. The dispatch ledger interface has an offline static-display fallback and never requires a digital identity to view.

N2 Fail-safe Test Yard

失效安全测试院

Courtyard + Station S02 S03

Opportunity
Create a public window into system testing and human takeover.

Spatial action
An everyday public court that uses movable boundaries during tests; an outer gallery lets visitors observe safely; a Courtyard + Station prototype.

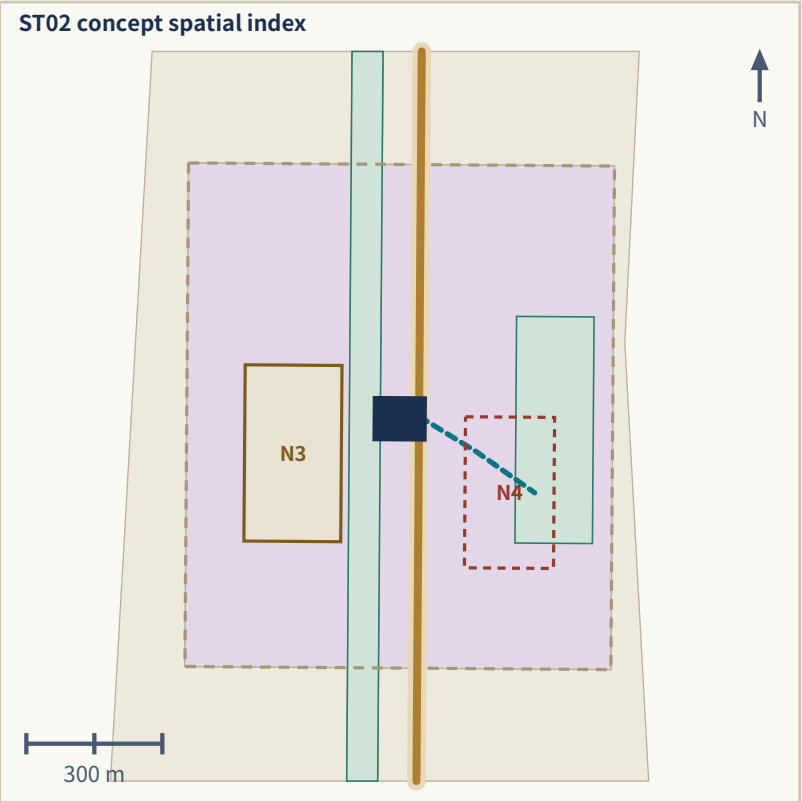
Operating modes
Public court or booked test yard by day; ordinary open space at night; shelter and gathering during extreme weather.

Users
Evaluation teams, regulators, visitors, and residents outside test periods.

Delivery inputs
Safety rules and permits, management responsibility, fire and structural review, and public, synthetic, or de-identified test data.

Deepening route
Complete siting, safety approval, facility, and operating design.

Manual / offline collaboration (declared geometry field)
Human takeover is the default action for any drill failure; printed instructions remain available on site. Staffed guidance is a design target during authorized drill/service periods only, subject to confirmed operator, with no digital step required.



ST02 • Verification and incubation

Beijing AI Origin Community

Origin Commons

PROV-KEY-002 • announced approx. 104.3 ha • recomputed 104.3 ha

Positioning

Validation and incubation for a world-class AI ecosystem; public services receive community and university review before citywide use.

Spatial structure

Courtyard-centered small-scale network linking campus, park, and neighborhood; open-source honors line the slow-mobility spine.

Building renewal

Retention and light renovation, adding launch, talent, residential support, and open collaboration; reversible conversion of underused ground floors.

Mobility and public space

Campus–park stitching, night lighting, and safe routes home. Origin Commons is enterable and lingerable, with a staffed arrival-service interface calibrated through operating tests.

AI scenarios

01 Open-Source Launch Hall, 04 Newcomer 72-Hour Settling-In, 06 Jing-Zhang Heritage and AI Public Education, 09 Research-to-SME Conversion Test.

Deepening focus

Establish data and research authorization, coordinate ownership and community interests, and co-design testing and renovation through long-term participation.

Project hooks

HX-03 Origin Commons and open-source honour display; HX-06 spine accessibility continuity and night-safety upgrade

N3 Origin Courtyard

原点庭院

Courtyard + Gateway

S01 S06

Opportunity

Give universities, companies, and residents a shared in-between place for validation, learning, and everyday exchange.

Spatial action

An open semi-enclosed courtyard, accessible street-level service and display, reservable validation places, a permanent staffed desk, and linear open-source honors; a Courtyard + Gateway prototype.

Operating modes

Courses, validation, launches, and rest by day; lighting, seating, and callable help at night; open ground floors become cooling/warming rest points in extreme weather.

Users

Faculty, students, developers, residents, newcomers, parents, and children.

Delivery inputs

Campus access agreements, ground-floor use consent, ownership and maintenance, and resident participation channels.

Deepening route

Fix location, dimensions, building assessment, and co-operation plan.

Manual / offline collaboration (declared geometry field)

Printed handbook available on site; a staffed service desk is a design target during authorized service periods only, subject to confirmed operator and annual authorization, not a permanent-staffing claim. Content review and any showcase can be withdrawn without notice, no login required.

N4 Campus–Station Stitch

校—站缝合节点

Stitch + Station (research task)

S04 S09

Research task (design_only_closed_pending_study)

remains closed while transport, accessibility, and engineering studies compare overpass, underpass, at-grade reconfiguration, shelter, end spaces, detours, and service arrangements.

Opportunity

Shorten detours between campuses, parks, and rail stations, especially at night and in rain or snow.

Spatial intent

A continuous step-free connection with gathering space, shelter, seats, lighting, and the connector alignment ; a Stitch + Station prototype. Professional comparison selects the crossing form.

Future operation

School and work trips by day; visible safe routes at night; alternate routes during extreme weather; staffed guidance aligned with the confirmed operator and authorized service windows.

Users

Faculty, students, commuters, night workers, parents, wheelchair users, and people with limited mobility.

Delivery inputs

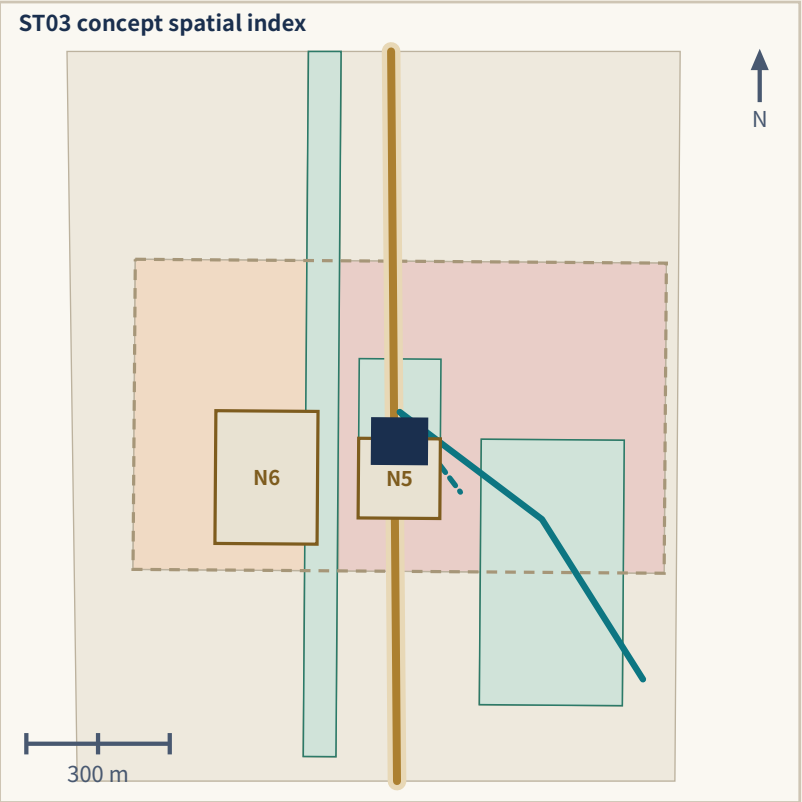
Transport, accessibility, and engineering studies; rail and road safety; station entrances; ownership and utilities at both ends.

Deepening route

Complete option selection, alignment, end-space design, and opening procedure after the studies.

Manual / offline collaboration (declared geometry field)

The research brief includes route continuity, offline wayfinding and a safe alternative under disruption. N4 remains design_only_closed_pending_study until transport, accessibility and engineering studies confirm the feasible authorized scheme.



ST03 • Launch and translation
Dazhongsi AI Industry Cluster

Axis Launch Station
PROV-KEY-003 • announced approx. 72.0 ha • recomputed 72.0 ha

Positioning
Launch and conversion area for intelligence-native business; the axis’ s public, market, and international interface.

Spatial structure
Market-led, with four-quadrant pedestrian continuity and ground-floor services around rail stations; launch and cultural content follow the walking network.

Building renewal
Retention and ground-floor activation, compound use of planned green space, and calibrated new or added space after planning and ownership confirmation.

Mobility and public space
Rail integration and four-quadrant walking connectivity, co-designed with transport and utilities. The launch station hosts the annual timetable and remains an everyday urban living room.

AI scenarios
05 Barrier-Free Continuous Passage, 07 Safe Route Home for Night Workers, 03 Network/Power Fail-safe, 10 Climate Digital-Twin Operations Test.

Deepening focus
Complete intersection safety, utilities, ownership, maintenance, and international content-rights procedures.

Project hooks
HX-05 Dazhongsi four-quadrant pedestrian connection; HX-07 Axis Timetable and annual open test season

N5 Four-Quadrant Crossing

四象限过街
Gateway + Station S05 S07

Opportunity
Integrate walking, transfers, waiting, and rest at all four corners of a large intersection.

Spatial action
Continuous step-free surfaces, sheltered waiting areas, drop-off, bicycle parking, and station entrances arranged as one Gateway + Station prototype; transport simulation selects the crossing form.

Operating modes
Commuting and interchange by day; lighting, visibility, and callable help at night; shelter and guided alternate routes during rain, snow, ponding, or ice.

Users
Transit riders, workers, visitors, older residents, wheelchair users, and cyclists.

Delivery inputs
Intersection safety and traffic review, red lines and utilities, station connections, ownership, and management.

Deepening route
Complete traffic simulation, crossing option, station integration, and municipal design.

Manual / offline collaboration (declared geometry field)
A non-digital, continuous ground-level route with no app or digital ticket required to cross is a design target, subject to professional transport/accessibility confirmation, an authorized service window, safety conditions, and annual authorization; no existing route is claimed. Staffed guidance is a design target during authorized service periods only, subject to confirmed operator and safety conditions, not a permanent-staffing claim.

N6 New Axis Launch Market

新轴发布市集
Market + Courtyard S03 S10

Opportunity
Let product launch, procurement, culture, and daily street life share one year-round space.

Spatial action
A permanent light canopy frames an open street used as market, rest, and display space; detachable stalls and panels transform it during launch season; a Market + Courtyard prototype.

Operating modes
Market, display, meetings, and rest by day; seating and small events at night; shade and rain protection in extreme weather, with temporary elements stored for strong wind or snow.

Users
Residents, workers, SMEs, start-ups, visitors, international guests, and vendors.

Delivery inputs
Ownership and use consent, event permit and safety plan, canopy structure and fire review, rights clearance and reuse of temporary elements.

Deepening route
Complete siting, dimensioning, structure, tenant mix, and everyday operation.

Manual / offline collaboration (declared geometry field)
Canopy and seating structure (design target) remains usable with power/network down; a printed contingency plan covers removal of temporary fixtures. Staffed presence during events is a design target subject to confirmed operator and authorized service periods, not a permanent-staffing claim.

Ten AI scenario cards sharing one rights boundary (human review, offline alternative, exit)

S01 Open-source Release Hall Origin Community	S02 Safety-governance Sandbox Zhongzhiyuan	S03 Network/power failure fail-safe Stations and gateways, whole axis	S04 Newcomer 72-hour settling Origin Community, gateways	S05 Continuous accessible movement Entire spine
S06 Jing-Zhang heritage and AI public education Heritage spine, Origin Community	S07 Safe night route for shift workers Spine and stations	S08 Haidian–Ulanqab green compute dispatch test Zhongzhiyuan AI industry test scenario	S09 Research-to-SME product translation test Origin Community, Dazhongsi AI industry test scenario	S10 Climate digital-twin operations test Whole axis, Qinghe–Xiaoyue AI industry test scenario

Governance duties: separation of public service, technical operation and independent audit

Duty	Scope and coordination interface
1 Public-interest trustee	Maintains free, continuous and accessible public space with physical service entry, reviewing impacts on basic service
2 Technical operator	Runs daily scenarios, dispatch, maintenance and incident response; performance is evaluated independently
3 Independent auditor	Verifies ledger definitions, energy/carbon/water metering and privacy enforcement; issues public audit opinions
4 Residents' and users' council	Represents residents, older people, mobility-limited users, night workers and students and initiates review and improvement
5 Haidian–Ulanqab agreement working group	Negotiates task tiers, metering and incident responsibility, implemented through a signed public project agreement
6 Independent appeals and mediation body	Receives complaints, mediates disputes and supervises suspension and exit

Reversible authorization cycle and shared card rules

90-day standards and no-personal-data drill (concept)
90-day standards rehearsal: establish rules, definitions, process and ledger structure with public, synthetic or de-identified data.

One-year reversible public pilot (concept)
One-year reversible public pilot: run at selected nodes and continuously adjust facilities, rules and service from evidence.

Annual renew, modify, or stop cycle (concept)
Annual audit and renew / modify / stop decision: independent audit supports transparent evidence-led decisions.

Shared rules: every card must have a usable non-digital alternative under network, power or system failure, and every card must let a user opt out without losing basic public service. S08, S09 and S10 are the three AI industry test scenarios; until real operating data exist the ledger is published as a test specification only, with no fabricated measurements.

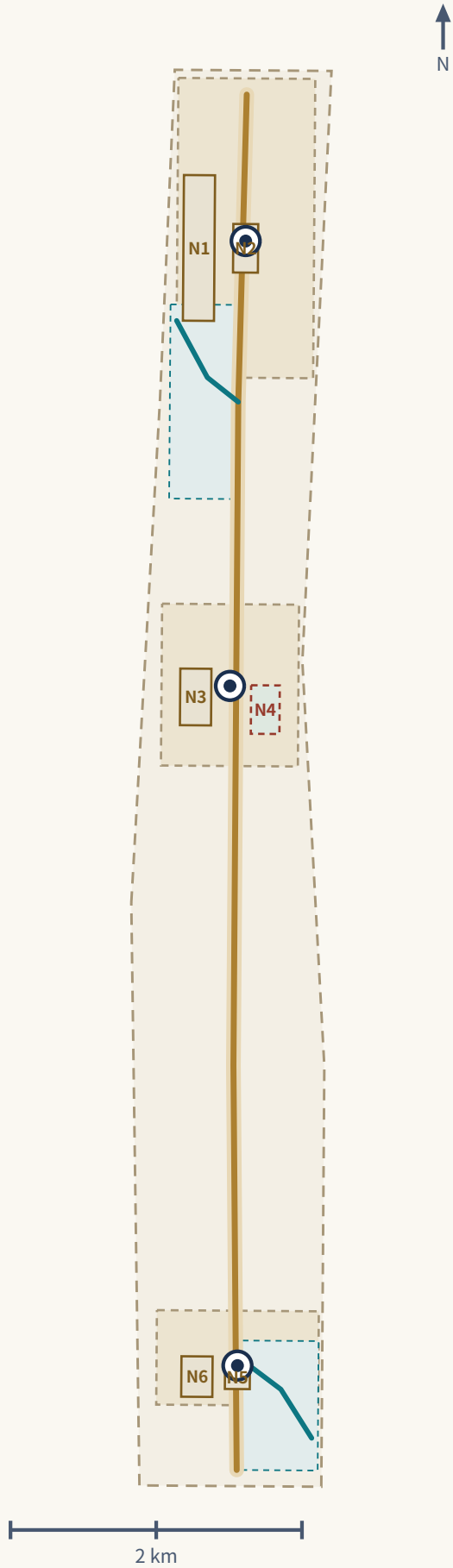
Renewal project portfolio for professional development

ID	Project	Type	Key dependencies	Deepening focus
HX-01	Spine walking/cycling gap stitching and continuous shade	Public space / mobility	Road right-of-way, space under bridges, traffic review	Use gap surveys and traffic simulation to select ring-road crossing solutions
HX-02	Qinghe–Xiaoyue blue-green cross-link and detention	Blue-green space	River blue line, ecological and flood conditions	Calibrate detention capacity with hydrology and engineering data
HX-03	Origin Commons and open-source honour display	Urban renewal / culture	Campus boundary, ownership, ground-floor uses	Use a co-creation charter to link campus, firms and communities
HX-04	Compute Exchange and public ledger interface	New infrastructure / public service	Energy, safety, operator and the Haidian–Ulanqab agreement	Use the project agreement to integrate real ledger data
HX-05	Dazhongsi four-quadrant pedestrian connection	Rail integration / active travel	Rail station, intersection, utilities	Select the engineering scheme through multi-agency option comparison
HX-06	Spine accessibility continuity and night-safety upgrade	Public space	Condition survey, lighting and maintenance body	Use existing-condition surveys to set continuity and night-service baselines
HX-07	Axis Timetable and annual open test season	Operations / brand	Public-space permits, event safety, rights clearance	Let the Open Operations Consortium establish an annual budget and durable program

Phasing: rules → pilots → stations → network → annual review

The near term establishes data and rights rules and launches reversible public-space pilots, the ledger and test specifications. The medium term builds the three stations after professional conditions are confirmed. The long term extends the lateral network and institutionalizes annual review. The open call's 100 days produce the submission; delivery then advances through the three phases.

Phasing layers (concept)



Three reversible authorization phases

Feature	Phase	Content	Declared flags
PHASE-001	10-day standards and no-personal-data drill (concept)	90-day standards rehearsal: establish rules, definitions, process and ledger structure with public, synthetic or de-identified data.	reversible=True personal_data_allowed=False
PHASE-002	One-year reversible public pilot (concept)	One-year reversible public pilot: run at selected nodes and continuously adjust facilities, rules and service from evidence.	reversible=True personal_data_allowed=False
PHASE-003	Annual renew, modify, or stop cycle (concept)	Annual audit and renew / modify / stop decision: independent audit supports transparent evidence-led decisions.	reversible=True personal_data_allowed=False

Long-term operations and global events (agent.6)

Annual events
The Axis Timetable is one calendar: the open test season (cards S08–S10), the open-source release season (S01), the heritage and public-education season (S06) and the annual ledger review; intensity follows season and climate.

Scenario operation
The ten cards open through apply, assess, trial, review and exit; any card can be terminated without affecting basic public service.

Attraction and translation
Auditable service performance (translation projects, exit rate, evaluation reports, participating teams) replaces capital-scale narrative; no investment, funding, policy or fiscal commitment is written in.

Fourteen synchronized spatial/design metrics (recalculated vs display value)

Metric	Machine value (metrics.json)	Display value	Meaning and limit
site_area_sqm	11412825.386 m²	approx. 11.4 km²	Working-boundary area; update when the official boundary is integrated
building_footprint_area_sqm	35851.926 m²	approx. 35,900 m²	Sum of three concept intervention envelopes, expressing the renewal action area
green_ratio	0.113434	approx. 11.3%	Union of six green concept surfaces ÷ provisional boundary area
public_space_ratio	0.044371	approx. 4.4%	Union of N1–N6 ÷ provisional boundary area
conceptual_mobility_length_m	11437.434 m	approx. 11.4 km	Sum of five concept slow-mobility links, checking structural completeness
key_area_count	3	3	Key areas
station_count	3	3	Station envelopes
wing_count	2	2	Wings
prototype_node_count	6	6	Prototype nodes
scenario_count	10	10	Operating scenarios
industry_test_count	3	3	Industry test scenarios
phase_count	3	3	Phases
declared_reversible_prototype_ratio	1.0	100% (declaration basis)	Submission/documentation coverage; built outcomes enter pilot and annual ledger
declared_manual_fallback_documentation_ratio	1.0	100% (declaration basis)	Submission/documentation coverage; built outcomes enter pilot and annual ledger

Fourteen metrics synchronize the current concept structure and display; official boundary, plan, engineering and operating inputs update them by tier.

Evidence chain: geometry → formula → metric → proposal citation

Geometry evidence	Formula	Metric
geometry/site_boundary.geojson#SITE-001	polygon_area(SITE-001) @EPSG:4548	site_area_sqm
geometry/green_space.geojson#GREEN-* ×6	union_area(GREEN-*) ÷ site_area_sqm	green_ratio
geometry/public_space.geojson#N1…N6	union_area(N1..N6) ÷ site_area_sqm	public_space_ratio
geometry/roads.geojson#ROAD-* ×5	Σ line_length(ROAD-*) @EPSG:4548	conceptual_mobility_length_m
geometry/buildings.geojson#BLDG-ST01/02/03-001	Σ polygon_area(BLDG-*)	building_footprint_area_sqm

Verifiable design targets: establish baselines, definitions and pilot data

Design target	Control quantity and formula	Value
Rest / cooling point coverage	Walking distance from any point on the spine to the nearest available point	baseline stage
Night availability of rest points	Open hours / night-period hours	baseline stage
Spine accessibility continuity	Accessible continuous length / total spine length	baseline stage
Failure recovery time	Time from degradation to recovery or completed human takeover	baseline stage
Staff arrival and help-response time	Time from call to staff arrival or answer	baseline stage
Fault repair duration	Time from fault registration to completed repair	baseline stage

Pathways for 23 professional and operating metrics

Plans and building surveys set development controls; accessibility, lighting and footfall surveys set public-service baselines; the Haidian–Ulanqab agreement and operating ledger set capacity, service and resource metrics.

The two declared_* metrics record proposal/documentation coverage for six node cards; pilot and annual ledger evidence evaluates built outcomes.